

Data Collection and Ingestion

Overview

This section describes the three stages that form the foundation of the knowledge base used across all retrieval experiments: dataset exploration, abstract retrieval from PubMed, and document ingestion into vector databases. These stages were executed in sequence and represent the complete preprocessing pipeline before any retrieval or generation experiment was run.

1. Dataset: PubMedQA

Source and Structure

The dataset used in this study is PubMedQA, a biomedical question answering benchmark originally proposed for research on clinical and scientific reasoning. The labeled subset, which is the version used here, consists of 1,000 samples drawn from published PubMed literature. Each sample contains a research question derived from the title of a biomedical paper, a structured context (comprising between one and nine supporting passage snippets from the original abstract), a long-form expert answer, and a final decision label classifying whether the evidence supports the answer as yes, no, or maybe.

The dataset was loaded from HuggingFace Datasets using the `bigbio/pubmed_qa` identifier with the `pubmed_qa_labeled_fold0_source` subset configuration.

Label Distribution

The 1,000 labeled samples are distributed as follows across the three decision categories:

Label	Count
yes	552
no	338
maybe	110

This imbalance reflects the natural tendency of PubMedQA questions to be framed such that the evidence leans affirmatively. The maybe class, representing ambiguous or inconclusive findings, is the smallest and arguably the most challenging for any QA system.

Context Structure

Each question in the dataset is paired with a context object containing a list of passage snippets (referred to as contexts), labels indicating the rhetorical role of each passage (such as BACKGROUND or RESULTS), and a list of Medical Subject Headings (MeSH terms). The number of context snippets per question has a mean of 3.36 and ranges from 1 to 9, with the vast majority of questions (720 out of 1,000) having exactly three supporting passages.

MeSH Term Analysis

MeSH terms provide a controlled vocabulary for categorising biomedical literature. Across the full dataset, 3,408 unique MeSH terms are represented. The most frequent terms reflect the general clinical nature of the corpus: Humans appears in 959 of the 1,000 records, followed by Female (785) and Male (703). Population-level descriptors such as Middle Aged (542) and Adult (492) are also highly prevalent. More specialised terms such as Myocardial Infarction, Stroke, and Obesity appear in the mid-frequency range (between 5 and 20 occurrences), indicating that the dataset spans a wide range of biomedical domains rather than being concentrated in any single specialty.

A MeSH lookup table mapping each PubMed ID to its associated MeSH terms was saved as `data/processed/mesh_lookup.json`. This mapping was later used to enrich each document chunk with domain metadata during ingestion.

2. Abstract Retrieval from PubMed

Approach

Although the PubMedQA dataset includes short context snippets for each question, these snippets represent only a portion of the full published abstract. For the purposes of building a retrieval corpus, the complete abstracts were fetched directly from PubMed using the NCBI E-utilities API (the `efetch` endpoint). This ensures that the knowledge base contains full-length abstracts rather than the pre-truncated snippets available in the dataset itself.

Each abstract was retrieved in plain text format and saved as an individual file at `data/raw/{pubid}.txt`, where pubid corresponds to the PubMed identifier of the paper. The download process includes an idempotency check: if a file already exists at the target path, it is skipped. This makes it safe to resume interrupted runs without re-fetching previously downloaded abstracts.

A one-second delay was enforced between each API call in compliance with NCBI's recommended rate limit for unauthenticated access. All 1,000 abstracts were downloaded successfully.

Verification

Following the download, the output directory was checked against the expected set of PubMed IDs. All 1,000 abstracts were confirmed to be present, with no missing entries.

3. Ingestion Pipeline

Parsing and Cleaning

The raw `.txt` files retrieved from PubMed carry formatting artefacts introduced by the NCBI export format. Each file begins with a citation header containing the journal reference, article title, author list, and affiliations, and ends with a footer section containing identifiers such as DOI, PMID, and PMCID. Some files also include a `Publisher:` block with a translated version of the abstract in a non-English language.

Two cleaning operations were applied before chunking:

Multilingual detection. Any abstract containing a non-English `Publisher:` translation block was identified using the `langdetect` library. A 400-character excerpt beginning at the `Publisher:` marker was sampled and its language detected. If the detected language was not English, the abstract was excluded from the knowledge base. One abstract (PubMed ID 24666444) was removed on this basis, leaving 999 documents eligible for ingestion.

Abstract body extraction. For each eligible file, the PubMed citation header and trailing footer were stripped to retain only the scientific content of the abstract. The extraction logic identifies the start of the abstract body by scanning for structured section headings (such as `BACKGROUND:`, `METHODS:`, or `RESULTS:`), an explicit `Abstract` heading, or, in the case of unstructured abstracts, the first sufficiently long paragraph occurring after the author and affiliation block. Everything before this start point and everything after the first footer marker is discarded.

Chunking

The cleaned abstract body text was split into overlapping chunks using `LangChain's RecursiveCharacterTextSplitter`. The splitter was configured with a chunk size of 800 characters and operates by attempting to break at paragraph, sentence, and word boundaries in that order of preference. This produces relatively self-contained chunks that tend to correspond to complete sentences or structured sections, which is important for retrieval quality in a biomedical setting where individual sentences can carry dense clinical information.

Each chunk was assigned a document ID of the form `{pubid}_chunk{n}` and was enriched with metadata including the originating PubMed ID, the source file path, the chunk index within the abstract, the character count, and the list of MeSH terms associated with the parent abstract.

In total, 2,849 chunks were produced from the 999 eligible abstracts. The chunked corpus was serialised to `data/knowledge_base/knowledge_base.json` as a list of records containing the chunk text and all metadata fields. This snapshot serves as the authoritative corpus for all downstream experiments.

Vector Database Ingestion

The same set of 2,849 chunks was ingested into four vector databases to allow cross-backend comparison in later retrieval experiments:

ChromaDB is a local, file-backed vector store persisted to disk under `vectorstores/minilm_pubmed_chromadb`. It supports cosine similarity search natively and is the primary backend used in retrieval experiments.

Qdrant is a local persistent collection stored under `vectorstores/minilm_pubmed_qdrant`. Vectors are indexed with cosine distance, and the collection was created programmatically with an embedding dimension inferred from the embedding model.

LanceDB stores chunks in a columnar Arrow-based format at `vectorstores/lancedb`. Each record includes the embedded vector alongside the text and metadata fields.

FAISS uses an in-memory flat index saved to disk as `index.faiss` and `index.pkl` under `vectorstores/minilm_pubmed_faiss`. Unlike the other stores, FAISS relies on its own internal integer indices and does not require explicit document IDs.

A fifth store, Weaviate, was also populated via a cloud-hosted cluster for exploratory comparison, though its use is not the focus of the main experimental results.

All four stores were verified to contain exactly 2,849 chunks following ingestion.

Embedding Model

All embeddings were generated using `sentence-transformers/all-MiniLM-L6-v2`, a compact 384-dimensional sentence embedding model. This model was selected for its balance of computational efficiency and semantic quality on short, factual text. The same model instance was used consistently across ingestion and retrieval to ensure that query vectors and document vectors inhabit the same embedding space.

Summary

The preprocessing pipeline takes the PubMedQA labeled dataset as its starting point, fetches the corresponding full abstracts from PubMed, applies language filtering and structural cleaning, and produces a chunked corpus of 2,849 passages stored across multiple vector databases. This corpus forms the shared knowledge base against which all retrieval-augmented generation experiments in this study are evaluated.